the fact that all 65 cards were PCI Express-based is testament to that!

Let's not waste any more time and get to the testing.

HIGH-END GPUS

The cards based around these graphics processors represent the cream of the bunch—the high flyers, the elite of the pixel-crunching monsters! And let's not forget the fact that they're also the most expensive!

The Elite
NVIDIA GeForce 7950GX2, 7900GTX
ATI X1950XTX, X1900XTX, X1900XT

Features

The five chipsets in this category represent the crème de la crème of graphics cards, and, currently, are the fastest GPUs on the planet. We received eight cards in this sub-category. Let's take a look at the architecture of these cards.

One look at the specifications alongside will tell you that the big boys aren't fooling around—mighty impressive! The 7950 GX2 is basically a dual-GPU solution, and actually has two interconnected PCBs (each with a GPU onboard). These PCBs are mated together by four hexahedral bolts, and interconnected via an x48 PCI Express link. The 7950 GX2 chipset is the highest of the high-end cards, and each core gets a dedicated 512 MB of GDDR3 memory!

Our first impression was that the cooling solution (NVIDIA's own reference design) on the GX2 cards was a little too small, compared to the massive heat-pipe-based solution that the 7900GTX sports. Even the ATI chipsets come with more robust-looking coolers. This was confirmed during the tests: the GX2s ran hotter than the other NVIDIA cards. A peculiar occurrence was the extreme heat around the rear part of the second PCB (the one whose GPU is concealed). Even the fan is tiny, and it's not very fast; we feel a better cooling solution should have been incorporated on such a card.

A notch below is the GeForce 7900GTX. Based around a single G71 core, the 7900GTX packs in some 278 million transistors on-die. All the

Elite Chipset features and Specs

	NVIDIA			ATI		
Part Name	7950 GX2	7900 GTX	7800GTX 512	X1950XTX	X1900XTX	X1900XT
Chipset Designation	2x G71	G71	G70	R580+	R580	R580
Manufacturing Process (nm)	90	90	110	90	90	90
Core Clock Speed (MHz) (Manufacturer reference speeds)	2x 500	650	550	650	650	625
Onboard Memory (MB) / Type (Manufacturer reference)	2x 512 / GDDR3	512 / GDDR 3	512 / GDDR3	512 / GDDR4	512 / GDDR3	512 / GDDR3
Memory Speed (MHz) (Manufacturer reference speeds)	1200	1600	1700	2000	1550	1450
Memory bus width (bits)	256	256	256	256	256	256
Pixel Shader units	2x 24	24	24	48	48	48
Vertex Shader units	2x 8	8	8	8	8	8

How We Tested

Test Rig

CPU: 2.93 GHz Core 2 Duo X6800; Motherboard: Gigabyte DQ6 P965; RAM: 1 GB (512 x 2 MB) Kingston DDR2 800 MHz (4-4-4-12); Hard Drive: Hitachi T7K 250 SATA-II

Test Software

We divided the applications into two categories: real-world applications (games) and synthetic applications and benchmarks.

The games were divided into two further categories on the basis of the API (Application Programming Interface) used—DirectX and OpenGL games.

DirectX Games

F.E.A.R. (First Encounter Assault Recon): A stunner of a game from Monolith that brings even the latest (and greatest) cards to their knees, with frame rates you can count on your fingers! Atmospheric and dark is how we describe this game. The eye-candy is amazing, the graphics mind boggling, and any graphics card test isn't quite complete without the cards being hammered by this game.

Far Cry: Another game that utilises some spectacular effects; the water and landscape effects of *Far Cry* have to bee experienced to be believed. Although not as gritty and realistic as *F.E.A.R*, *Far Cry* is still one of the best-looking first person shooters around.

Chaos Theory: The third in the *Splinter Cell* series, this game is characterised by some great lighting (or lack of it), and very real and interactive environments. One of the very best tactical shooters available, in terms of gameplay and visual effects.

OpenGL Games

DOOM 3: No longer a new title, but still one of the most shadow intensive games out there, this one will thrill even as it chills. Moreover, *DOOM 3* is very scalable—perfect to test the wide range of cards we received.

Prey: Based around a modified and enhanced version of the *DOOM 3* engine, *Prey* manages to look as good as (if not better) than *DOOM 3* and even *Quake 4* (another *DOOM 3* engine-based shooter).

We used the latest patches, if any, that were available at the time for the games.

Synthetic Software

We used the latest versions of 3D Mark 05 and 06, which are pretty standard gear for any graphics card benchmarking. They stress the pixel processing power of all cards, in particular 3D Mark 06, which, as you'll see, is very demanding even on the best of cards.

The Categories

This time around, we decided to divide the cards based on chipsets instead of by manufacturers. The cards were split into the regular three categories: High-End, Mid-Range, and Entry-Level. However, this distinction was made on the basis of what chipset was used, instead of by prices. We further divided the High-End cards into sub-categories on the basis of chipsets—the top-of-the-line (Elite Gamer) cards and other high-end (Enthusiast Gamer) cards.

The prices of the graphics cards also conveniently fell into place, since cards based on a particular chipset, from different manufacturers, have very marginal price differences.

We decided that the high-end chipsets would comprise NVIDIA's 7950GX2, 7900GTX/GT and 7800GTX/GT series, and ATI's X1900 XTX/XT/GT and X1800 XT/XL/GTO series.

The mid-range chipsets were ATI's X850/X800 and X1600 series, and NVIDIA's 6800 and 7600 series.

Entry-level cards ran on NVIDIA's 7300, 6600 and 6200 series of chipsets, and ATI's X1300 and X700/X600/X300 series.